Leonardo Maglanoc

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Profile

Robotics & AI Engineer (M.Sc., TUM) with experience in robot learning, machine learning, and 3D perception. Developed C++ robotics software, PyTorch simulators, and LLM-based manipulation pipelines.

Experience

Robotics Master's Thesis Collaboration – Foundation Robotics & TUM MIRMI 10/2025 – 05/2026
Munich, full-time

- Researched dexterous manipulation with San Francisco-based humanoid startup and TUM MIRMI.
- Developed Lang2Tool in Python using LLM tool calling through the OpenAI API to translate language instructions into geometric motion primitives and SE(3) trajectories.
- Integrated these motion primitives with an RL policy controlling a Sharpa Hand in Isaac Lab.

AI/ML Research Intern – BMW Group Research 02/2025 – 08/2025
Munich, full-time

- Co-developed a PyTorch simulator with IBM for oscillatory neural networks (ONNs) using Neural ODEs.
- Led a tactile interaction study with 10 participants; trained and evaluated ONNs for human-vehicle interfaces.

Robotics Research Assistant – TUM 04/2023 – 05/2024
Munich, part-time

- Contributed to SaRA-Shield, a reachability-based safety framework for AI-driven robot manipulation (e.g., reinforcement learning); implemented kinematics, dynamics, and trajectory planning in C++.
- Co-authored an IEEE Transactions on Robotics paper; the framework improved task completion speed by 15.9% over the best-performing baseline.

Selected Projects

Applied AI & Coding Agent Projects 09/2026

- Rapidly prototyped interactive AI demos with coding agents, including a browser-based G1 humanoid parkour simulation (Demo).
- Developed EasyBuy, an agentic shopping assistant, at the Hack-Nation hackathon; invited to Venture Lab.

SO-101 Robot Learning 08/2026

- Built a LeRobot pipeline on a physical SO-101: teleoperation, dataset collection, ACT/SmolVLA training for a pick-and-place task.

LiDAR-Visual-Inertial SLAM – Guided Research, TUM 04/2024 – 09/2024

- Implemented and evaluated LiDAR feature extraction and voxel filtering in C++ and ROS within a multi-sensor SLAM system for drone navigation, using real HILTI 2022 sensor data.
- Used GDB and Valgrind to debug the C++ SLAM pipeline and investigate memory issues.

Education

M.Sc. Robotics, Cognition, Intelligence – TUM 04/2023 – 06/2026
Thesis: *Lang2Tool: Language-Guided SE(3) Tool-Motion Generation for Dexterous Manipulation*.

B.Sc. Informatics – TUM 09/2019 – 03/2023
Thesis: *Provably Safe Human-Robot Interaction for Manipulation using Power and Force Limiting*.

Publication

J. Thumm, J. Balletshofer, **L. Maglanoc**, L. Muschal, M. Althoff. "A general safety framework for autonomous manipulation in human environments." *IEEE Transactions on Robotics*, 2026.

Technical Skills

Robotics: Manipulation · Motion Planning & Control · 3D Perception & SLAM · Navigation
AI & ML: Robot Learning (IL/RL) · Foundation Models · Agentic AI · Computer Vision
Tools: Python · C++ · PyTorch · ROS2 · LeRobot · Isaac Lab · MuJoCo · Linux · Git · Docker

Languages

German and Norwegian (native); English (C2); French (B1).